European Business Review

# Enhanced supply chain resilience in global supply chains: a multi-dimensional capability framework

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Received 26 March 2026
Revised 28 April 2026
20 May 2026
Accepted 13 July 2026

## Abstract

**Purpose** – Global supply chain interdependencies increase firm's disruption exposure, intensifying scholarly and managerial interest in resilience. Despite previously identified resilience activities, limited understanding exists of how capabilities are developed, aligned and enacted across organizational and inter-organizational contexts. Therefore, this study aims to investigate how firms build supply chain resilience through coordinated capability configurations.

**Design/methodology/approach** – A systematic literature review using Scopus and Web of Science, synthesizing 52 peer-reviewed articles. Extracted data was descriptively and thematically analyzed, identifying resilience-related activities and capabilities and examining their distribution across organizational functions, supply chain levels and theoretical perspectives.

**Findings** – Supply chain resilience arises from operational, relational and transformational capabilities across internal, supply chain and ecosystem levels. This study introduces a three-dimensional framework linking who enacts resilience, where it is targeted, and it's strategic justification, while proposing a typology of resilience orientations spanning single dimensional, to fully integrated resilience approaches.

**Research limitations/implications** – This study advances a configurational and capability-based understanding of supply chain resilience by integrating the resource-based view, relational view and dynamic capabilities view. The proposed framework and typology provide a foundation for cumulative theory development and future empirical research on resilience capability alignment.

**Practical implications** – The findings of this study offer a structured approach for diagnosing and developing resilience strategies. Through a multidimensional perspective, firms can better align capabilities across functions and supply chain levels and balance efficiency and redundancy in dynamic environments.

**Originality/value** – This study moves beyond descriptive reviews of resilience activities by developing an integrative framework and typology explaining how resilience capabilities are systematically configured and enacted in global supply chains.

**Keywords** Global supply chain resilience, Resource-based view, Relational view, Dynamic capabilities view, Systematic literature review

**Paper type** Literature review

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European Business Review
Emerald Publishing Limited
0955-534X
DOI 10.1108/EBR-03-2026-0135

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## 1. Introduction

Supply chain resilience has become a central theme in both research and practice, reflecting the increasing vulnerability of globally interconnected supply networks. Contemporary supply chains extend across multiple firms and geographies, creating complex interdependencies that expose them to diverse risks and disruptions (OECD, 2024; Wieland and Durach, 2021). Disruptions such as the COVID-19 pandemic, the war in Ukraine, the Suez Canal blockage and trade and geopolitical tensions in the Strait of Hormuz have illustrated how vulnerabilities can ripple through entire networks. These events have intensified scholarly and managerial attention to resilience as a defining capability of supply chains, commonly understood as the ability to withstand, adapt to and recover from disruptions while ideally emerging in a stronger position (Christopher and Peck, 2004; Hohenstein *et al.*, 2015; Tukamuhabwa *et al.*, 2015). The surge in publications regarding supply chain resilience after 2019 (Castillo, 2023; Chowdhury *et al.*, 2021; Rahman *et al.*, 2022) underscores the growing urgency of this discussion and positions resilience as a critical lens for analyzing contemporary supply chain management.

Against this backdrop, this study contributes to the ongoing discourse on supply chain resilience by examining how resilience is developed and sustained in global supply chains. While practices such as multisourcing, network diversification and ecosystem partnerships (Gartner, 2020) have been identified as mechanisms for strengthening readiness before disruptions, enhancing responsiveness during disruptions and supporting recovery in their aftermath (Rahman *et al.*, 2022), questions remain regarding how resilience capabilities are developed, enacted, shared and refined over time in dynamic environments (Stadtfeld and Gruchmann, 2024; Wieland and Durach, 2021). Accordingly, this study reviews and synthesizes the extant literature to advance understanding not only of which capabilities support disruption resilience but also how they are developed and combined through organizational activities and collaborative, knowledge-driven approaches extending across supply chains. The research is guided by the following questions:

*RQ1*. What capabilities enable enhanced disruption resilience in global supply chains?

*RQ2*. How are resilience capabilities organized and structured within and across firms and supply chains?

*RQ3*. How do firms combine and align resilience capabilities to sustain resilience in dynamic environments?

This study adopts a configurational and theory-informed multidimensional perspective on supply chain resilience, focusing on patterns of capability integration within and across firms. Rather than seeking to establish common generic prescriptions or performance benchmarks (Datta, 2017; Ivanov *et al.*, 2023; Scholten *et al.*, 2025), the analysis emphasizes patterns of capability integration and strategic coherence across organizational and inter-organizational contexts. The scope of this study is, therefore, limited to empirical research and resilience-related practices that are documented in peer-reviewed literature. This analytical focus advances conceptual understanding of resilience as a multidimensional configuration shaped by the interaction of organizational functions, structural arrangements and strategic rationales, while acknowledging that such configurations remain context dependent.

## 2. Theoretical framework

Supply chain resilience generally refers to the ability of firms to navigate and counteract disturbances and return to their original state or move to a new, more desirable state after

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disruption (Christopher and Peck, 2004; Hohenstein *et al.*, 2015; Tukamuhabwa *et al.*, 2015; Wieland and Durach, 2021).

### *2.1 Antecedents of supply chain resilience*

Firms faced, and coped with, supply chain disruptions before the concept of supply chain resilience was introduced. Supply chain resilience could be argued as having its origin in 2004 (Christopher and Peck, 2004), where the first definition was proposed (Castillo, 2023). Supply chain resilience should be viewed as a continuum in which multiple capabilities are continuously enhanced to reduce disturbance vulnerability (Aslam *et al.*, 2020; Stadtfeld and Gruchmann, 2024). Therefore, concepts that pre-date supply chain resilience such as total quality management, supply chain risk management (Chowdhury and Quaddus, 2016; Tang, 2006) and robustness (Aslam *et al.*, 2020) could be viewed as antecedents of and prerequisites for achieving supply chain resilience (Aslam *et al.*, 2020).

Efficiency-oriented capabilities, reducing redundancies, such as lean practices, may be counterproductive (Aslam *et al.*, 2020; Christopher and Peck, 2004; Tukamuhabwa *et al.*, 2015), as they can increase firm vulnerability during disruptions. Where tensions can rise as resilience aims to reduce disturbance vulnerability, entailing a degree of redundancy (Riccardo *et al.*, 2021). Firms, therefore, need capabilities that support both efficiency and resilience, maintaining competitiveness while reducing vulnerabilities (Aslam *et al.*, 2020).

### *2.2 Supply chain readiness, response and recovery*

Supply chain resilience hinges on developing readiness before, response during and recovery after disturbance (Chowdhury *et al.*, 2021; Christopher and Peck, 2004; Hohenstein *et al.*, 2015; Orlando *et al.*, 2022; Peck, 2006; Sheffi and Rice, 2005; Tukamuhabwa *et al.*, 2015). Where the degree of supply chain resilience depends on a firm's persistence and ability to cope with rapid change (Wieland and Durach, 2021).

Enhancing readiness, response and recovery capabilities reduces supply chain vulnerabilities (Chowdhury *et al.*, 2021), shortening the time required to regain normal operations after disruption (Tukamuhabwa *et al.*, 2015). The time between disruption impact and regained operations is visualized in the recovery triangle (Tukamuhabwa *et al.*, 2015). The model conceptualizes how firm performance before, during and after disturbance could be quantitively analyzed, as well as recovery time where performance is negatively impacted (Tukamuhabwa *et al.*, 2015; Figure 1).

The recovery triangle can be used for assessing a firm's degree of readiness, response and recovery (Tukamuhabwa *et al.*, 2015). However, extant literature lacks empirical evidence (Behzadi *et al.*, 2020; Castillo, 2023; Stentoft and Mikkelsen, 2024) of capabilities constituting, and contributing to, successful and long-term readiness, response and recovery.

### *2.3 Supply chain resilience across levels and organizational functions*

Commonly used levels to analyze supply chains are internal, supply chain and ecosystem (Eriksson and Svensson, 2015; Miemczyk *et al.*, 2012). Internal-level analysis focuses on capabilities enhancing single-firm resilience and are enacted internally. Supply chain-level analysis concerns resilience-enhancing capabilities developed in cooperation with external actors directly involved in inbound and outbound logistics processes, such as suppliers and customers at different tiers. Finally, ecosystem-level analysis addresses capabilities developed in cooperation with external actors who are not directly associated with the supply chain, such as governmental and regulatory entities.

Supply chain resilience is commonly analyzed from a firm perspective, whereas a broader analysis could highlight nuances (Yang *et al.*, 2025) contributing to a holistic understanding.

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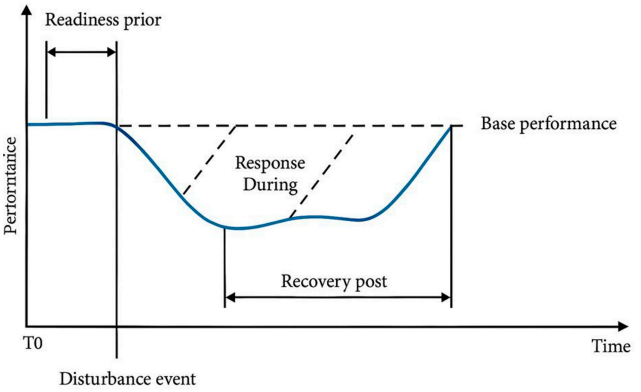



**Figure 1.** The recovery triangle (Tukamuhabwa *et al.*, 2015). Shaded area represents loss of performance as a consequence of disruption

Furthermore, supply chain resilience depends on interactions within firms, across supply chain actors and with ecosystem actors. This relates to a micro-, meso- and macro-level perspective of supply chain resilience (Zhao *et al.*, 2024), emphasizing that supply chain resilience is enhanced and maintained through interactions at several supply chain levels (Adobor, 2019).

Supply chain resilience needs to be analyzed across multiple organizational functions in which capabilities are built-up. Specifically, upper management, procurement, logistics, production and marketing and sales are examined. These functions are closely linked with inbound and outbound logistics processes and, therefore, to the broader supply chain. Resilience-enhancing capabilities are enacted through these functions, through internal, supply chain or ecosystem cooperation and interactions.

Capabilities aimed at enhancing supply chain resilience at the firm level focus on internal interaction within and across organizational functions (Pettit *et al.*, 2010; Van Den Adel *et al.*, 2023). These interactions allow resilience to permeate the firm, enabling efficient resource reallocation and process reconfiguration through multi-function cooperation and synchronization (Sá *et al.*, 2019; Tukamuhabwa *et al.*, 2015). This should be aligned with contextual characteristics and requirements. The emphasis on contextuality lies in the context-dependent nature of organizational activities (Fenton and Langley, 2011), implying that resilience-related capabilities and associated activities should be understood as inherently contextual.

### *2.4 Supply chain resilience a resource, dynamic and relational perspective*

The complex and multidimensional nature of supply chain resilience necessitate the integration of multiple theoretical perspectives (Shekarabi *et al.*, 2025). Accordingly, this study draws on Resource-Based View (RBV), Dynamic Capabilities View (DCV) and Relational View (RV).

RBV states that firms sustain competitive advantage through tangible and intangible resources (Wernerfelt, 1984) that are valuable, rare, inimitable and non-substitutable (Barney, 1991). This view recognizes resource homogeneity amongst firms (Eisenhardt and Martin, 2000; Hunt and Lambe, 2000; Peteraf, 1993; Tukamuhabwa *et al.*, 2015) within static business environments (Kero and Bogale, 2023). Despite this, an RBV perspective can

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be beneficial for enhanced supply chain resilience. By recognizing resource homomgeneity, this perspective highlights internal resource sharing (Tukamuhabwa *et al.*, 2015), enabling enhanced resilience through redundancy and internally mobile resources. Furthermore, strengthening firm resilience through strategic internal resources facilitates the strategic use of external resources for supply chain resilience (Long *et al.*, 2024; Shekarabi *et al.*, 2025).

DCV accounts for the dynamic nature of business environment and assumes heterogeneity amongst firms, in contrast to RBV (Eisenhardt and Martin, 2000). DCV conceptualizes sustained competitive advantage in highly technological (Teece *et al.*, 1997), rapidly changing (Eisenhardt and Martin, 2000), global and uncertain (Barreto, 2010) business environments arising from the ability to build, integrate and reconfigure internal and external resources (Shekarabi *et al.*, 2025; Teece *et al.*, 1997). Capabilities that enhance supply chain resilience are often deployed in dynamic business environments (Stadtfeld and Gruchmann, 2024; Wieland and Durach, 2021) in cooperation with external supply chain actors (Mandal *et al.*, 2016). Accordingly, the DCV provides a perspective that accounts for both internal and external resource use for enhancing supply chain resilience (Shekarabi *et al.*, 2025; Teece *et al.*, 1997). Dynamic capabilities can support supply chain resilience, for example through early disruption detection, resource mobilization for impact mitigation and process adaptation before, during and after disruptions (Stadtfeld and Gruchmann, 2024).

RV positions inter-organizational relationships as a source of competitive advantage (Dyer and Singh, 1998). From this perspective, firms develop competitiveness through relational processes supporting inter-organizational cooperation toward joint goals (Dyer and Singh, 1998; McCauley and Palus, 2021). RV is pivotal in enabling dynamic capabilities (Mandal *et al.*, 2016), leveraging external resources for competitive advantages (Teece *et al.*, 1997). Systematic development of relational capabilities through structured systems (Blackhurst *et al.*, 2011) and inter-organizational knowledge-sharing (Pu and Qiao, 2025) can enhance supply chain resilience in uncertain business environments (Wieland and Wallenburg, 2013). Moreover, RV facilitates access to external resources through inter-organizational relationships, thereby enhancing resilience (Mandal *et al.*, 2016).

## 3. Research method

Given the fragmented and multidisciplinary nature of research on supply chain resilience, a systematic literature review was adopted to ensure a comprehensive, transparent and replicable synthesis of prior empirical studies. Furthermore, the literature review laid the foundation for the proposed conceptual model presented. This approach enables the generation of insights in accordance with the study's purpose and research questions, while also providing an overview of extant knowledge in the field. The literature search followed a six-step process proposed by Durach *et al.* (2017), to collect and synthesize empirical evidence on how firms in global supply chains enhance resilience. The operationalization of this process is presented below. First, the focus on resilience-enhancing activities informed the formulation of the research questions (Step 1). Thereafter, the characteristics of the primary research were determined. In this process, inclusion/exclusion criteria were operationalized (Newbert, 2007; Figure 2). Included articles consisted of empirical studies were specific activities enacted for enhanced supply chain resilience within global supply chains were presented. Articles that were non-empirical and did not present specific activities enacted for enhanced supply chain resilience in global supply chains were excluded.

An initial pool of potentially relevant articles was retrieved using the search string ("supply chain" AND "resili*" AND ["practice*" OR "activit*" OR "practition*"]). The search included title, abstract and keywords of peer-reviewed journal articles in the databases Scopus and Web of Science. No other constraints were imposed, resulting in 3,576

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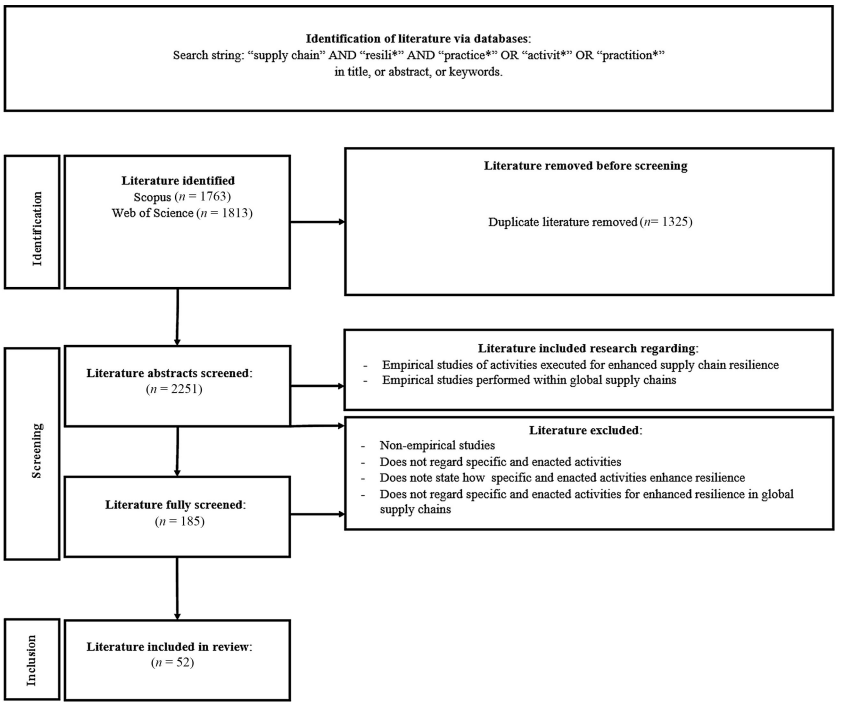



**Figure 2.** Visualization of achieving synthesis sample (Durach *et al.*, 2017)

potentially relevant entries (1,763 in Scopus and 1,813 in Web of Science). After removing 1,325 duplicates, 2,251 unique articles remained.

The search was conducted on December 31, 2025, alongside database alerts to ensure that newly published and relevant entries were subsequently screened, thereby maintaining the currency of the review (Step 3). The remaining 2,251 articles then underwent abstract screening to assess relevance to the study's purpose and research questions and quality criteria (Step 4). Thereafter, 185 articles were subjected to full-text screening for data extraction after abstract removal (Step 5). This process resulted in the inclusion of 52 articles (Figure 2).

The extracted data were synthesized using thematic analysis. This analysis followed the six step process by Braun and Clarke (2006; Table 1). The resulting themes and synthesized findings are presented in the chapter "Content analysis".

Data familiarization was conducted once synthesis sample had been established. All included articles were read in full to ensure substantive and empirical relevance. Relevant data in form of quotes were then highlighted, extracted and systematically organized using Microsoft Excel. In total, 458 quotes were extracted and structured. The remainder of the thematic analysis, from initial code generation to defining categories, was performed through seven iterative steps (Figure 3).

Figure 3 illustrates the analytical process described above and provides a detailed account of how each step was conducted. In Steps 1–3, extracted quotations were converted into first-level codes. Step 4 involved the development of second-level codes representing activities

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**Table 1.** Performing thematic analysis through six steps (Braun and Clarke, 2006)

| Thematic analysis steps | Corresponding activities |
|---|---|
| Step 1: Data familiarization | Relevant articles were read in full |
| Step 2: Initial code generation | Relevant data were highlighted, extracted and systematically organized |
| Step 3: Theme search | Extracted data were grouped based on semantic similarities to generate initial main categories |
| Step 4: Theme review | Main categories were reviewed in relation to the study's purpose and research questions |
| Step 5: Defining main categories | Main categories were examined and refined based on similarities among underlying codes |
| Step 6: Result synthesis | Synthesized results were presented in written form |

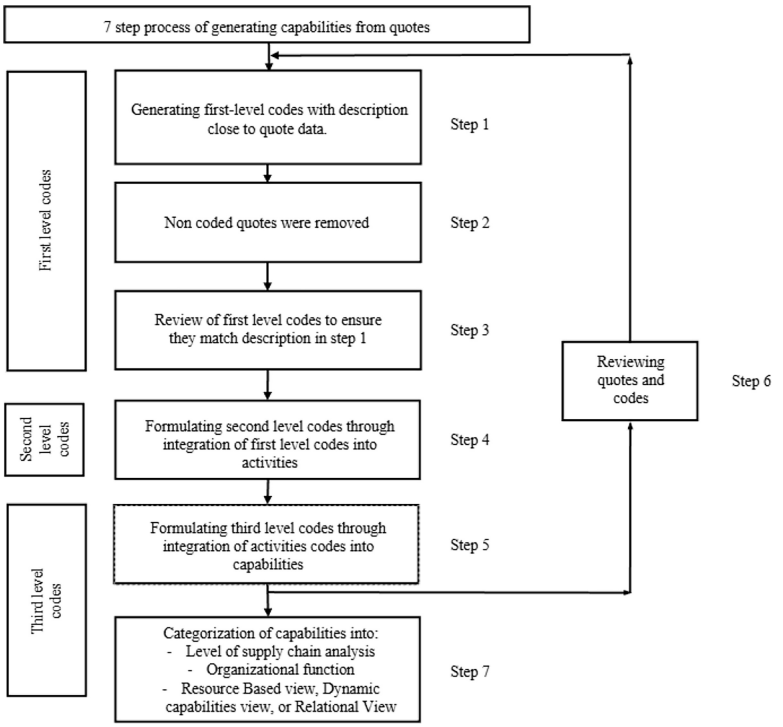



**Figure 3.** Illustration of performing seven iterative steps
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aimed at enhancing resilience in global supply chains. In Steps 5–6, these activities were integrated into third level codes representing capabilities that support supply chain resilience.

Step 7 focused on categorizing resilience capabilities in relation to the supply chain level they primarily address and the organizational functions in which they are enacted. In addition, the identified capabilities were categorized in relation to three organizational

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theories to develop a holistic understanding of how resilience is enhanced in global supply chains. The results were subsequently synthesized and presented in written form.

Figure 4 illustrates the coding structure, showing how first-level codes were generated from extracted quotations, clustered and interpreted into second-level codes and subsequently integrated into third-level codes aligned with the theoretical categories.

## 4. Descriptive analysis

Of the 52 synthesized articles, 40 (85%) have been published between 2020 and 2025. This increase in publications of supply chain resilience during this period is likely related to the global disruption caused by the COVID-19 pandemic (Figure 5).

The included articles were published across 37 academic journals. Most journals are represented by one single article, whereas nine journals are represented by between two and five publications (Figure 6).

An analysis of the research methods used in the synthesized articles indicates that case studies are the most common approach, accounting for 32% of the articles included. It is followed by survey and interview studies, which represent 31% and 23% of the articles, respectively. Mixed methods, modeling and Delphi studies together account for the remaining 14% (Figure 7).

## 5. Content analysis

The chapter presents capabilities that enhance resilience in global supply chains. These capabilities are examined in relation to supply chain level of analysis and organizational function. The analytical approach was not predetermined but emerged through an iterative process (Dubois and Gadde, 2002), reflecting a parsimonious alignment between theory and empirical material. The process identified the RBV, RV and DCV as central theoretical perspectives for explaining supply chain resilience.

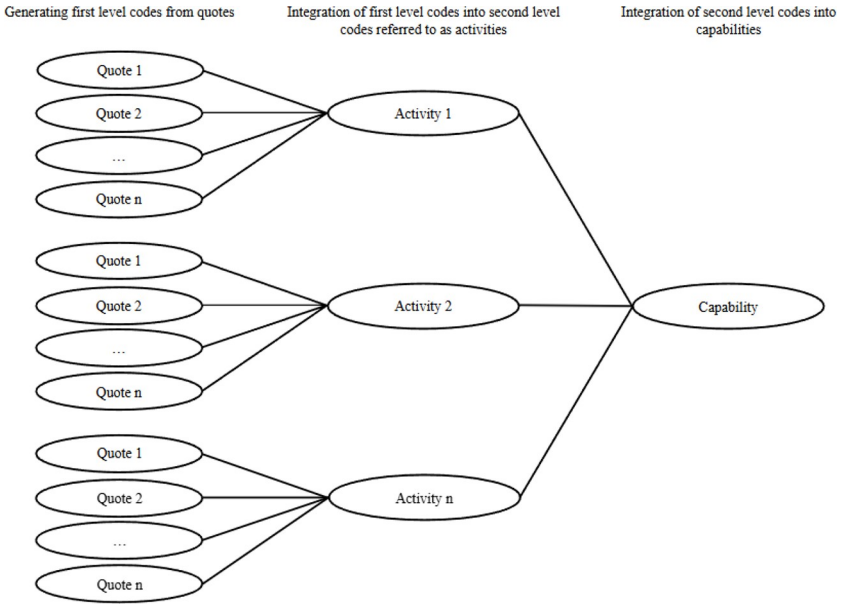



**Figure 4.** Illustration of coding flow
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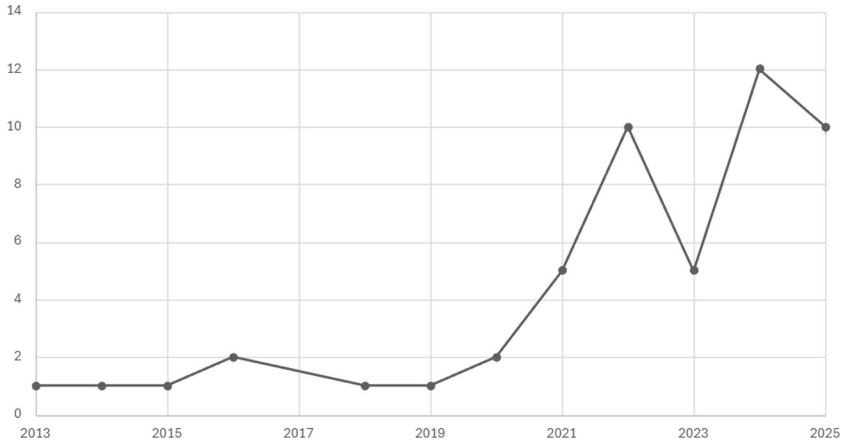



**Figure 5.** Publication count per year
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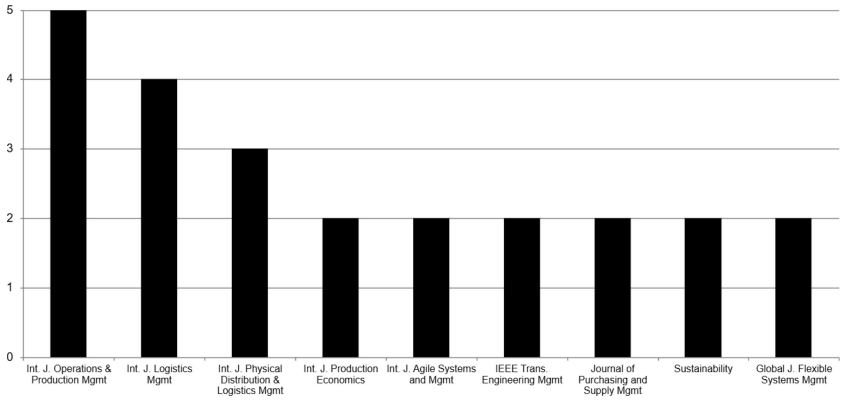



**Figure 6.** Distribution of journals represented by two or publications
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### *5.1 Capabilities*

Firms engage in multiple activities to address disruptions, many of which aim at similar outcomes. These activities can, therefore, be grouped into capabilities that support resilience in global supply chains (Table 2). Such capabilities are enacted by different actors, operating in varying contexts and are grounded in distinct theoretical logics. Accordingly, the analysis organizes capabilities in relation to supply chain level and organizational functions, providing a framework tailored to the present research context, in line with how Dubois and Gadde (2002) view the role of a framework.

Within this framework, three types of capabilities are distinguished. RBV-based capabilities are conceptualized as "operational" capabilities, reflecting the use of internal resources to maintain operations and sustaining competitive advantages. RV-based capabilities are understood as "relational" capabilities, emphasizing the development and

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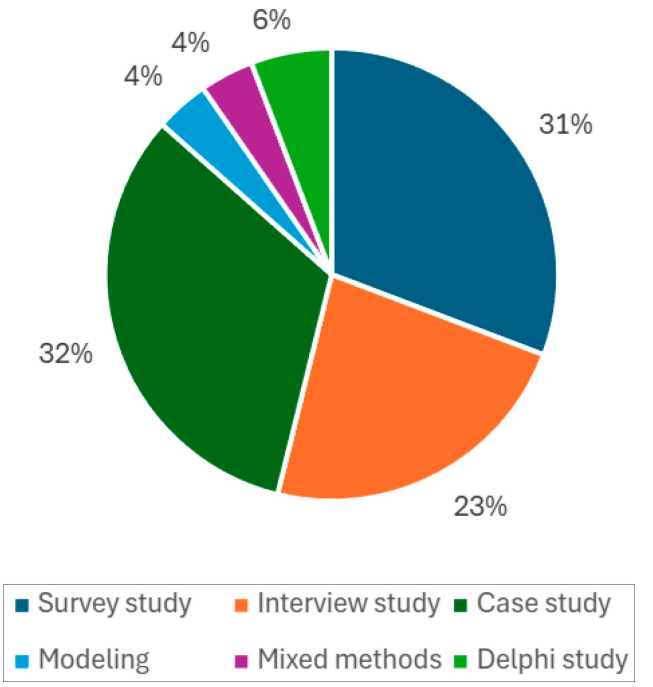



**Figure 7.** Distribution of methods between articles
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leveraging of interorganizational relationships to enhance resilience. DCV-based capabilities are conceptualized as "transformational" capabilities, reflecting firms' ability to adapt internal and external processes in response to disruptive environmental changes.

*5.2 Categorization based on level of analysis*

The first analytical categorization of capabilities is based on levels of supply chain analysis (Miemczyk *et al.*, 2012). This categorization reflects the primary level at which each capability is directed. Accordingly, three levels are distinguished: internal, supply chain and ecosystem (Table 3, Figure 8).

Capabilities that build internal resilience comprise activities conducted within the focal firm's sphere of control to enhance its ability to cope with disruptions. Such activities include ensuring continuity of operations through buffering of critical components (Bastas and Garza-Reyes, 2022; Carvalho *et al.*, 2013; Vanany *et al.*, 2024). Firms may also leverage existing capabilities and processes to support portfolio diversification (Peters *et al.*, 2023) by repurposing materials and resources (Dittfeld *et al.*, 2022; Okorie *et al.*, 2020). Together, these activities enable firms to adjust internally to disruptions by enhancing process flexibility while maintaining alignment between demand and supply (Cherrafi *et al.*, 2022; Nagariya *et al.*, 2024; Scholten *et al.*, 2025; Vanany *et al.*, 2024).

Capabilities that build resilience at the supply chain level involve activities conducted in collaboration with supply chain partners, such as suppliers or 3PL providers. A key example is multiple sourcing (Carvalho *et al.*, 2013; Day *et al.*, 2024; Majumdar and Srivastava,

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**Table 2.** Identified capabilities

| Capability | Definition | Activities | Sources |
|---|---|---|---|
| Inventory management | Ability to deliberately manage the movement and positioning of inventory | • Products moved to avoid becoming obsolete<br>• Strategic buffering | Bastas and Garza-Reyes (2022), Carvalho *et al.* (2013), Cherrafi *et al.* (2022), Chowdhury *et al.* (2022), Cohen *et al.* (2022), Colicchia *et al.* (2011), Dittfeld *et al.* (2022), Elzarka (2021), Küffner *et al.* (2022), Lopez-Ridaura *et al.* (2021), Mohd Rashid *et al.* (2014), Peters *et al.* (2023), Scholten and Schilder (2015) and Vanany *et al.* (2024) |
| Supply chain control | Ability to strengthen oversight of procurement and supplier relations | • Centralization of supply functions<br>• Securing suitable contracts<br>• Increase supply chain visibility<br>• Reducing supply chain complexity | Alvarenga *et al.* (2023), Bastas and Garza-Reyes (2022), Belhadi *et al.* (2024), Betto and Garengo (2023), Cherrafi *et al.* (2022), Cohen *et al.* (2022), Dabhilkar *et al.* (2016), Day *et al.* (2024), Dittfeld *et al.* (2022, 2021), Domingos *et al.* (2024, Enz *et al.* (2024), Küffner *et al.* (2022), Mandal *et al.* (2016), Mohd Rashid *et al.* (2014), Nagariya *et al.* (2024), Norrman and Wieland (2020), Paul *et al.* (2023), Peters *et al.* (2023), Scholten and Schilder (2015), Silva and Ruel (2022), Spieske *et al.* (2022) and Trabucco and De Giovanni (2021) |
| Disruption-specific adaptation | Ability to allow rapid process adjustments | • Implementation of disaster protocols<br>• quickly and suitable response to specific disruptions | Dabhilkar *et al.* (2016), Dittfeld *et al.* (2021), Enz *et al.* (2024), Nagariya *et al.* (2024), Paul *et al.* (2023), Peters *et al.* (2023), Vanany *et al.* (2024) and Yang and Hsu (2018) |
| Stakeholder coordination | Ability to engage external actors | • Rapid communication<br>• Mutual support through external stakeholders | Alvarenga *et al.* (2023), Bastas and Garza-Reyes (2022), Dabhilkar *et al.* (2016), Dittfeld *et al.* (2022), Mohd Rashid *et al.* (2014), Moyo *et al.* (2026), Norrman and Wieland (2020), Paul *et al.* (2023), Scholten and Schilder (2015), Vanany *et al.* (2024) and Zighan *et al.* (2024) |
| Supply assurance | Ability to secure operational continuity | • Building redundancy in supply<br>• Relocation of supply chain activities<br>• Maintaining flexibility with suppliers | Alvarenga *et al.* (2023), Bastas and Garza-Reyes (2022), Belhadi *et al.* (2024), Bugvi *et al.* (2024), Carvalho *et al.* (2013), Cherrafi *et al.* (2022), Cohen *et al.* (2022), Colicchia *et al.* (2011), Dabhilkar *et al.* (2016), Day *et al.* (2024), Dittfeld *et al.* (2022, 2021), Domingos *et al.* (2024), Echefaj *et al.* (2024), Elzarka (2021), Enz *et al.* (2024), Farrukh and Sajjad (2025), Kaeo-Tad *et al.* (2021), Kouam (2025), Küffner *et al.* (2022), Lopez-Ridaura *et al.* (2021), Majumdar and Srivastava (2025), Mandal *et al.* (2016), Mohd Rashid *et al.* (2014), Nagariya *et al.* (2024), Norrman and Wieland (2020), Ocicka *et al.* (2022), Okorie *et al.* (2020), Peters *et al.* (2023), |

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**Table 2.** Continued

| Capability | Definition | Activities | Sources |
|---|---|---|---|
| | | | Scholten and Schilder (2015), Spieske *et al.* (2022), Trabucco and De Giovanni (2021) and Vanany *et al.* (2024) |
| Market intelligence | Ability to understand and anticipate change | • Understanding demand patterns<br>• Basing decisions on demand patterns | Cohen *et al.* (2022), Nagariya *et al.* (2024) and Vanany *et al.* (2024) |
| Process efficiency | Ability to maintain efficient processes | • Efficient utilization of inputs<br>• Minimize waste | Bastas and Garza-Reyes (2022), Cherrafi *et al.* (2022), Cohen *et al.* (2022), Dittfeld *et al.* (2022, 2021), Elzarka (2021), Kaeo-Tad *et al.* (2021), Küffner *et al.* (2022), Nagariya *et al.* (2024), Okorie *et al.* (2020), Peters *et al.* (2023), Scholten *et al.* (2025), Scholten and Schilder (2015) and Zighan *et al.* (2024) |
| Process flexibility | Ability to adapt operations to changing conditions or requirements | • Flexibility in processes to allow adaptation<br>• Matching supply capacity with demand | Bastas and Garza-Reyes (2022), Cherrafi *et al.* (2022), Cohen *et al.* (2022), Dittfeld *et al.* (2022, 2021), Domingos *et al.* (2024), Elzarka (2021), Küffner *et al.* (2022), Madhavika *et al.* (2024), Majumdar and Srivastava (2025), Mohd Rashid *et al.* (2014), Nagariya *et al.* (2024), Norrman and Wieland (2020), Okorie *et al.* (2020), Paul *et al.* (2023), Scholten and Schilder (2015), Spieske *et al.* (2022), Vanany *et al.* (2024) and Zhang and Zhao (2025) |
| Lead time reduction | Ability to ensure a quick supply chain | • Acceleration of in<br>• bound logistics processes<br>• Acceleration of out<br>• bound logistics processes | Elzarka (2021), Mandal *et al.* (2016), Nagariya *et al.* (2024) and Norrman and Wieland (2020) |
| Supplier development | Ability to jointly enhance supplier resilience | • Increasing supplier ability to maintain continuous operations<br>• Enabling continuous and reliable supply | Betto and Garengo (2023), Cohen *et al.* (2022), Day *et al.* (2024), Dittfeld *et al.* (2022), Elzarka (2021), Mohd Rashid *et al.* (2014), Norrman and Wieland (2020) and Silva and Ruel (2022) |
| Financial resource control | Ability to manage and leverage financial flow | • Systematic management of internal financial flow<br>• Systematic management of external financial flows | Cohen *et al.* (2022), Nagariya *et al.* (2024), Tanveer *et al.* (2025) and Zhang and Zhao (2025) |
| Supply chain integration | Ability to strengthen the coordination with | • Integration of supply chain actors<br>• Cooperation with supply chain actors | Belhadi *et al.* (2024), Bugvi *et al.* (2024), Cherrafi *et al.* (2022), Cho *et al.* (2025), Cohen *et al.* (2022), Dabhilkar *et al.* (2016), Day *et al.* (2024), Dittfeld *et al.* |

(*continued*)

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**Table 2.** Continued

| Capability | Definition | Activities | Sources |
|---|---|---|---|
| | supply chain partners | | (2022), Domingos *et al.* (2024), Elzarka (2021), Enz *et al.* (2024), Küffner *et al.* (2022), Madhavika *et al.* (2024), Mandal *et al.* (2016), Mohd Rashid *et al.* (2014), Nagariya *et al.* (2024), Paul *et al.* (2023), Peters *et al.* (2023), Scholten and Schilder (2015), Silva and Ruel (2022), Spieske *et al.* (2022), Tao *et al.* (2025), Trabucco and De Giovanni (2021), Yang and Hsu (2018) and Zighan *et al.* (2024) |
| Supply localization | Ability to reduce reliance on distant suppliers | • Source nearby<br>• Reducing distance to supplier | Cohen *et al.* (2022), Kaeo-Tad *et al.* (2021), Mohd Rashid *et al.* (2014) and Sharma *et al.* (2025) |
| Transport flexibility | Ability to maintain resilient logistics | • Use of multiple transportation modes<br>• Ensuring transportation mode adaptability<br>• Transportation through 3PL | Alvarenga *et al.* (2023), Carvalho *et al.* (2013), Colicchia *et al.* (2011), Enz *et al.* (2024), Küffner *et al.* (2022), Lopez-Ridaura *et al.* (2021), Majumdar and Srivastava (2025), Nandy *et al.* (2025), Norrman and Wieland (2020), Okorie *et al.* (2020), Scholten and Schilder (2015), Spieske *et al.* (2022) and Vanany *et al.* (2024) |
| Market responsiveness | Ability to sustain and increase revenue | • Create more time to sell products<br>• Discounts to sell more<br>• Offer security to customers<br>• Keep customers engaged<br>• Use of multiple sales channels | Alvarenga *et al.* (2023), Bastas and Garza-Reyes (2022), Cherrafi *et al.* (2022), Chowdhury *et al.* (2022), Cohen *et al.* (2022), Dittfeld *et al.* (2021), Elzarka (2021), Enz *et al.* (2024), Kaeo-Tad *et al.* (2021), Kouam (2025), Lopez-Ridaura *et al.* (2021), Paul *et al.* (2023), Scholten and Schilder (2015), Spieske *et al.* (2022) and Trabucco and De Giovanni (2021) |
| Workforce management | Ability to balance cost efficiency, flexibility, and employee well-being | • Temporary lowering salary<br>• Flexible allocation of employees<br>• Maintaining expected work environment and employee safety | Bastas and Garza-Reyes (2022), Cherrafi *et al.* (2022), Cohen *et al.* (2022), Dabhilkar *et al.* (2016), Kaeo-Tad *et al.* (2021), Kouam (2025), Madhavika *et al.* (2024), Nagariya *et al.* (2024), Sharma *et al.* (2025) and Vanany *et al.* (2024) |

**Source(s):** Authors' own work

2025; Vanany *et al.*, 2024) which creates supply redundancy (Echefaj *et al.*, 2024) and enhances firms' ability to meet demand. Furthermore, firms may strengthen collaboration with supply chain partners (Mandal *et al.*, 2016). Such collaboration can enable the outsourcing of critical activities such as manufacturing (Alvarenga *et al.*, 2023) and buffer replenishment (Mohd Rashid *et al.*, 2014). In addition, close geographical proximity to supply chain partners may further support resilience and can be prioritized, when possible

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**Table 3.** Categorization of capabilities based on level of analysis

| Internal resilience | Supply chain resilience | Ecosystem resilience |
|---|---|---|
| Inventory management | Supply chain control | Market intelligence |
| Disruption-specific adaptation | Supply assurance | Stakeholder coordination |
| Process efficiency | Lead time reduction | |
| Process flexibility | Supplier development | |
| Market responsiveness | Financial resource control | |
| Workforce management | Supply chain integration | |
| | Supply localization | |
| | Transport flexibility | |

**Source(s):** Authors' own work

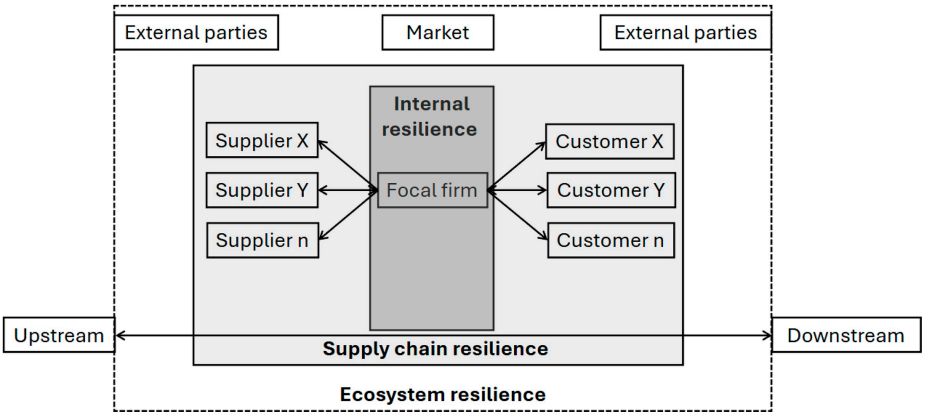



**Figure 8.** Visualization of categorization based on level of analysis
**Source:** Authors' own work

(Kaeo-Tad *et al.*, 2021; Peters *et al.*, 2023), for example through local sourcing (Cohen *et al.*, 2022; Kaeo-Tad *et al.*, 2021; Mohd Rashid *et al.*, 2014).

Capabilities that build ecosystem-level resilience extend beyond the immediate supply chain and involve engagement with broader institutional and market actors. These include responding to market shifts through forecasting and monitoring activities (Vanany *et al.*, 2024), which support early disruption detection (Nagariya *et al.*, 2024). Such activities also facilitate more rapid communication with external stakeholders, which is critical for disruption mitigation (Scholten and Schilder, 2015). In addition, ecosystem-level capabilities include financial support from governmental agencies to sustain operations (Bastas and Garza-Reyes, 2022) and cooperation with research institutions to support capability development (Paul *et al.*, 2023).

This level-based categorization demonstrates that supply chain resilience is built through a multi-level approach. Even when individual firms seek to enhance resilience, their effort must address internal, supply chain and ecosystem levels simultaneously. Conceptualizing resilience in this way also enables the application of complementary theoretical perspectives, such as principal agent theory (Jensen and Meckling, 1998), game theory and supply chain orchestration (Ermini *et al.*, 2024).

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*5.3 Categorization based on organizational function*

The second categorization model is based on what organizational function of the focal firm that is mainly responsible for enacting resilience capabilities. The categories are upper management, procurement, logistics (inbound/outbound), production and marketing and sales (Table 4; Figure 9).

Upper management holds primary responsibility and authority for adaptively managing strategic decisions (Dittfeld *et al.*, 2021) to ensure continuity of operations and enhance resilience. This includes implementing firm-wide risk management protocols (Peters *et al.*, 2023) and facilitating communication and collaboration with supply chain partners (Dabhilkar *et al.*, 2016; Zighan *et al.*, 2024) and ecosystem stakeholders, such as governmental agencies (Vanany *et al.*, 2024). Such engagement may enable access to

**Table 4.** Categorization of capabilities based on organizational functions

| Upper management | Procurement | Logistics | Production | Marketing and sales |
|---|---|---|---|---|
| Disruption-specific adaptation | Supply chain control | Lead time reduction | Inventory management | Market intelligence |
| Stakeholder coordination | Supply assurance | Transport flexibility | Process efficiency | Market responsiveness |
| Financial resource control | Supplier development | | Process flexibility | |
| Workforce management | Supply chain integration | | | |
| | Supply localization | | | |

**Source(s):** Authors' own work

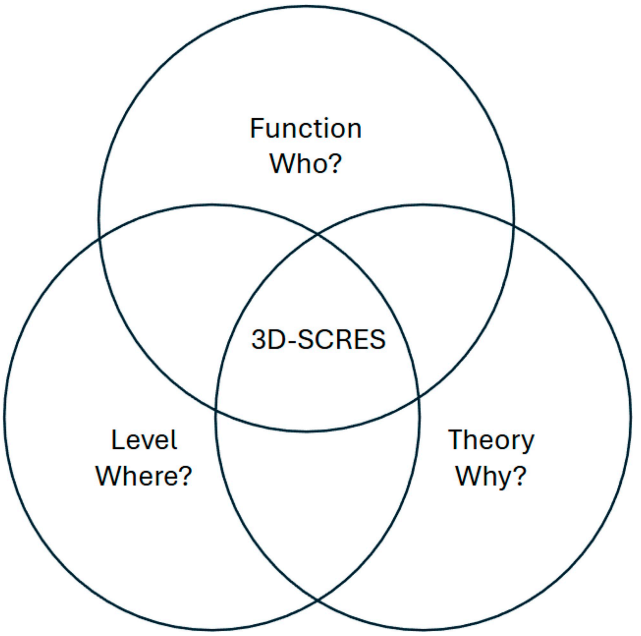



**Figure 9.** Three-dimensional supply chain resilience
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financial support that sustains operations during disruptions (Bastas and Garza-Reyes, 2022). In addition, upper management plays a central role in workforce management, including the provision of safe working conditions during crises such as COVID-19 (Bastas and Garza-Reyes, 2022; Cherrafi *et al.*, 2022).

The procurement function contributes to resilience by ensuring supply assurance through practices such as multi-sourcing (Cherrafi *et al.*, 2022; Cohen *et al.*, 2022; Majumdar and Srivastava, 2025; Silva and Ruel, 2022), local sourcing (Cohen *et al.*, 2022; Kaeo-Tad *et al.*, 2021; Mohd Rashid *et al.*, 2014) and strategic buffering (Bastas and Garza-Reyes, 2022). These activities are supported by close communication and relations with suppliers, enabled through integration and collaboration (Belhadi *et al.*, 2024; Bugvi *et al.*, 2024; Cohen *et al.*, 2022) and strategic supplier localization (Day *et al.*, 2024; Farrukh and Sajjad, 2025). Procurement also facilitates supplier development through systematic monitoring (Dittfeld *et al.*, 2022; Mohd Rashid *et al.*, 2014).

The logistics focuses on maintaining efficient material flows from inbound deliveries to warehousing and outbound distribution. This supports supply chain velocity (Mandal *et al.*, 2016) and can be achieved through lead time reduction (Elzarka, 2021), the integration of appropriate technologies (Nagariya *et al.*, 2024) and the prioritization of efficient and geographically proximate suppliers (Norrman and Wieland, 2020; Elzarka, 2021). Resilience is further strengthened through multimodal transportation (Colicchia *et al.*, 2011; Spieske *et al.*, 2022) and the use of alternative transportation routes (Carvalho *et al.*, 2013; Norrman and Wieland, 2020). The use of 3PL providers (Vanany *et al.*, 2024) also enhances flexibility and supports convergent material flows (Alvarenga *et al.*, 2023).

The production function enhances resilience by minimizing waste through practices such as First Expiry First Out (Chowdhury *et al.*, 2022) and by avoiding overproduction (Cherrafi *et al.*, 2022), thereby contributing to cost efficiency (Cherrafi *et al.*, 2022; Cohen *et al.*, 2022). At the same time, resilience depends on maintaining efficient and flexible production processes, that can be adjusted in response to disruptions (Dittfeld *et al.*, 2021; Spieske *et al.*, 2022; Zhang and Zhao, 2025). Efficiency is supported by the effective use of existing resources (Peters *et al.*, 2023) and inputs (Dittfeld *et al.*, 2022; Nagariya *et al.*, 2024), while flexibility enables firms to better align demand and supply (Dittfeld *et al.*, 2021; Elzarka, 2021; Nagariya *et al.*, 2024; Vanany *et al.*, 2024). Production flexibility can be fostered through process standardization (Cohen *et al.*, 2022), investments aimed at reducing production stoppages (Dittfeld *et al.*, 2021), the use of multiple production sites (Cohen *et al.*, 2022; Norrman and Wieland, 2020) and postponement strategies (Scholten *et al.*, 2025).

The market and sales function contribute to resilience by supporting market awareness and managing and maintaining sales performance. Forecasting activities (Vanany *et al.*, 2024) enhance market sensitivity (Nagariya *et al.*, 2024) and enable firms to adjust product portfolios in line with demand and profitability considerations (Cohen *et al.*, 2022). This function also supports continuity during disruptions through market diversification (Bastas and Garza-Reyes, 2022; Kaeo-Tad *et al.*, 2021; Lopez-Ridaura *et al.*, 2021), product diversification (Bastas and Garza-Reyes, 2022) and the development of multiple sales channels (Chowdhury *et al.*, 2022; Enz *et al.*, 2024). In addition, sales resilience may be supported through targeted discounts (Chowdhury *et al.*, 2022), customer financing arrangements (Cohen *et al.*, 2022) and customer integration initiatives (Enz *et al.*, 2024; Spieske *et al.*, 2022).

The categorization based on organizational functions demonstrates that supply chain resilience is not achieved through the efforts of a single department. Instead, it emerges from coordinated contributions across multiple functions. Consistent with the supply chain management model proposed by Cooper *et al.* (1997), resilience should, therefore, be

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understood as a cross-functional and process-oriented organizational outcome that is embedded across organizational boundaries.

### *5.4 Mapping capabilities against theory*

Table 5 presents the previously identified capabilities mapped to the RBV (operational), RV (relational) and DCV (transformational).

Operational capabilities comprise internally oriented capabilities enacted across multiple organizational functions. In line with the RBV, these capabilities are grounded in the use of internal resources that are tangible or intangible (Wernerfelt, 1984) and that enable the creation of sustained competitive advantage (Barney, 1991; Peteraf, 1993; Wernerfelt, 1984). Through resource deployment, operational capabilities enhance resilience primarily at the internal and supply chain levels of analysis by supporting internal activities. This aligns with the RBV emphasis of firm-specific and immobile resource utilization (Barney, 1991; Peteraf, 1993). Sustaining competitive advantages depends on the strategic use of valuable, rare, inimitable and non-substitutable resources (Barney, 1991) to facilitate and maintain resilience-enhancing capabilities. Moreover, the effective deployment of internal resources may support access to external resources through cooperation with actors in the supply chain and broader ecosystem (Long *et al.*, 2024; Shekarabi *et al.*, 2025).

Enhancing resilience through the RV emphasizes access to external resources through systematic relationship development with supply chain and ecosystem actors (Mandal *et al.*, 2016). Relational capabilities can be leveraged to pursue mutual goals (McCauley and Palus, 2021), thereby strengthening resilience (Pu and Qiao, 2025). These capabilities are primarily enacted through organizational functions with external interfaces, such as upper management and procurement, and contribute to resilience across multiple levels of supply chain analysis. By maintaining and strengthening relationships with suppliers and ecosystem actors through interorganizational cooperation (Dyer and Singh, 1998; McCauley and Palus, 2021), firms enhance their capacity to respond to disruptions. Previous research shows that such relational capabilities support resilience during disruptive periods (Blackhurst *et al.*, 2011; Wieland and Wallenburg, 2013) and provide an essential foundation for enabling change through dynamic capabilities (Mandal *et al.*, 2016; Teece *et al.*, 1997).

Transformational capabilities correspond to the DCV and reflect a firm's ability to adapt to changing circumstances in global and uncertain business environments (Eisenhardt and Martin, 2000; Barreto, 2010). These capabilities are grounded in the integration and reconfiguration of internal and external resources (Shekarabi *et al.*, 2025; Teece *et al.*, 1997).

**Table 5.** Capabilities mapped against theory

| Operational (RBV) | Relational (RV) | Transformational (DCV) |
|---|---|---|
| Inventory management | Stakeholder coordination | Disruption specific adaptation |
| Supply chain control | Supplier development | Market intelligence |
| Process efficiency | Financial resource control | |
| Lead time reduction | Supply chain integration | |
| Supply localization | Market responsiveness | |
| Workforce management | | |
| Supply assurance | | |
| Process flexibility | | |
| Transport flexibility | | |

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Empirically, transformational change is primarily initiated though organizational functions such as upper management and marketing and sales, contributing to internal and ecosystem resilience. These capabilities enable firms to anticipate disruptions, mobilize resources for mitigation and reconfigure internal and external processes to sustain competitive advantage, in line with the DCV framework (Teece, 2007). Prior research indicates that transformational capabilities play a key role in enhancing resilience in dynamic business environments through the effective deployment of internal and external resources (Stadtfeld and Gruchmann, 2024; Mandal *et al.*, 2016).

The categorization demonstrates that integrating multiple theoretical perspectives enhances the understanding of resilience-related capabilities. The findings highlight the complexity of supply chain resilience and its dependence on multiple capabilities operating across different level of analysis (Ponomarov and Holcomb, 2009). Whereas previous research has often examined supply chain resilience within specific functional or thematic domains rather than from a holistic perspective (Azmi *et al.*, 2025; Shekarabi *et al.*, 2025), this study shows that a combined application of the RBV, RV and DCV is necessary to adequately explain the phenomenon. Notably, operational and relational capabilities were more prevalent than transformational capabilities. This pattern may reflect the tendency for resilience to be enacted primarily through reactive responses and generic mitigation strategies following disruptions (Datta, 2017; Scholten *et al.*, 2025; Ivanov *et al.*, 2023).

The categorizations presented above are fully analytically distinct and can, therefore, be integrated into a multidimensional coding framework. Integration enables simultaneous positioning of capabilities in relation to supply chain level, organizational function and theoretical perspective. For example, the capability "Supply chain control" is enacted at the supply chain level, through the procurement function, and can be classified as an operational capability. Such multidimensional positioning illustrates how resilience-related capabilities are distributed across supply chain levels, organizational functions and theoretical perspectives (Table 6).

## 6. Discussion

This study identified a diverse set of resilience-enhancing capabilities enacted across multiple organizational functions, supply chain levels and theoretical perspectives. The findings demonstrate that supply chain resilience emerges from the interaction of operational, relational and transformational capabilities distributed across internal, inter-organizational and ecosystem domains. Whereas prior research has often examined resilience within isolated functional or thematic areas, the present analysis adopts an integrative perspective that captures the multidimensional nature of resilience. Building on this empirical foundation, the following sections interpret the findings through established theoretical lenses and develop a three-dimensional framework that advances the understanding of how firms organize and enact resilience in global supply chains.

A central pattern emerging from the review is the predominance of operational and relational capabilities relative to transformational capabilities. This suggests that resilience is frequently pursued through stabilization-oriented measures such as redundancy, coordination and partner collaboration, whereas more transformative reconfiguration capabilities receive comparatively less emphasis. This imbalance indicates that supply chain resilience may often remain reactive rather than fully adaptive, potentially constraining long-term resilience renewal. This imbalance also suggests that long-term resilience may depend on not only maintaining operational and relational stability but also strengthening capability renewal through reconfiguration and adaptation. A second pattern is that resilience emerges as a distributed rather than centrally owned capability. The review shows that resilience-

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**Table 6.** Overview of capability categorization and theory mapping

| Capability | Categorization | | |
|---|---|---|---|
| | Level of analysis | Organizational function | Label |
| Inventory management | Internal resilience | Production | Operational (RBV) |
| Supply chain control | Supply chain resilience | Procurement | |
| Process efficiency | Internal resilience | Production | |
| Lead time reduction | Supply chain resilience | Logistics | |
| Supply localization | Supply chain resilience | Procurement | |
| Workforce management | Internal resilience | Upper management | |
| Supply assurance | Supply chain resilience | Procurement | |
| Process flexibility | Internal resilience | Production | |
| Transport flexibility | Supply chain resilience | Logistics | |
| Stakeholder coordination | Ecosystem resilience | Upper management | Relational (RV) |
| Supplier development | Supply chain resilience | Procurement | |
| Financial resource control | Supply chain resilience | Upper management | |
| Supply chain integration | Supply chain resilience | Procurement | |
| Market responsiveness | Internal resilience | Marketing and sales | |
| Disruption specific adaptation | Internal resilience | Upper management | Transformational DCV |
| Market intelligence | Ecosystem resilience | Marketing and sales | |

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enhancing activities are enacted across organizational functions, supply chain partners and ecosystem actors, suggesting that resilience is less a firm-level property than an interdependent coordination phenomenon. This also implies that resilience depends on not merely the presence of capabilities but also mechanisms for coordinating them across actors and functions. A third pattern is that resilience appears to depend less on the number of capabilities a firm possesses than on how those capabilities are aligned across functional, structural and strategic dimensions. This shifts attention from capability accumulation toward capability configuration, emphasizing the importance of multidimensional alignment in sustaining resilience.

### *6.1 Toward three-dimensional supply chain resilience (3D-SCRES)*

Adopting a multidimensional perspective enhances understanding of which capabilities are enacted through what activities, which capabilities are found at different organizational functions, where in the supply chain these capabilities aim to contribute to resilience and which theoretical perspective explains their relevance. Incorporating organizational theories as an additional analytical dimension provides deeper insight into resilience mechanisms and supports a more holistic understanding, which remains limited in much of the existing literature (Azmi *et al.*, 2025; Shekarabi *et al.*, 2025).

The findings further demonstrate that supply chain resilience cannot be enhanced through singular dimensions. The review also showed that individual resilience capabilities frequently span multiple dimensions simultaneously, suggesting that capabilities are rarely purely functional, structural or strategic in isolation. Instead, resilience emerges from the interaction of multiple capabilities across different supply chain levels and organizational functions. Consequently, the categorizations should not be viewed independently but as interconnected dimensions that collectively shape multidimensional resilience configurations.

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Within this framework, the capability represents *what* should be developed; organizational functions indicate *who* develops them; supply chain levels specify *where* their effects are directed; and theoretical perspectives explain *why* they are important. A comprehensive understanding of supply chain resilience, therefore, requires not only identifying relevant capabilities but also analyzing them within a three-dimensional supply chain resilience (3D-SCRES) framework that includes who, where and why.

It should also be emphasized that the activities underlying capability development are context specific (Fenton and Langley, 2011), implying that appropriate capability configurations may vary across settings. As this study focuses on global supply chains, the proposed dimensions are tailored to this context and should be interpreted as indicative rather than prescriptive, offering a structured basis for analyzing resilience from multiple complementary perspectives.

### *6.2 Typology of resilience orientations*

Building on the 3D-SCRES framework, firms can be classified according to the extent to which they integrate organizational functions (*who*), supply chain levels (*where*) and theoretical orientations (*why*) in their resilience efforts. This classification reveals three distinct resilience orientations: a *single-dimensional orientation*, characterized by a focus on one analytical dimension; a *partially integrated orientation*, reflecting the integration of two dimensions; and a *holistic integration orientation*, in which all three dimensions are systematically aligned. These resilience orientations should not be understood as fixed categories, but as potentially dynamic positions through which firms may evolve as capabilities are developed and aligned over time.

Firms exhibiting a *single-dimensional orientation* concentrate their resilience efforts on a single analytical dimension, organizational functions, supply chain level or theoretical rationale, while neglecting the remaining dimensions. Such a narrow focus limits the integration of operational, structural and strategic considerations, thereby constraining the development of sustained supply chain resilience. Although these firms may achieve localized improvements in specific areas, their resilience efforts tend to remain fragmented and vulnerable to systemic disruptions.

Firms adopting a *partially integrated orientation* combine two of the three analytical dimensions, resulting in more coordinated but still incomplete resilience strategies. Different forms of partial integration can be distinguished. Firms focusing on *who* and *where* may implement well-coordinated operational and structural measures across functions and supply chain levels. However, without a strong emphasis on *why*, such efforts risk lacking strategic direction and coherence. Consequently, capabilities may be developed that are operationally efficient but insufficiently aligned with long-term strategic objectives (Porter, 1996).

Firms emphasizing *who* and *why* integrate organizational functions with strategic intent, enabling internally aligned and purpose-driven resilience initiatives. Nevertheless, limited attention to *where* may restrict the extension of these capabilities across supply chain and ecosystem levels, reducing their effectiveness in managing inter-organizational disruptions. Similarly, firms combining *where* and *why* may develop strategically informed, network-oriented resilience practices, yet insufficient integration across internal functions can hinder effective implementation and coordination.

Firms exhibiting a holistic integration orientation systematically integrate organizational functions (*who*), supply chain levels (*where*) and theoretical rationales (*why*) in their resilience strategies. These firms align operational practices, inter-organizational structures and strategic intent within a coherent and mutually reinforcing framework. By simultaneously coordinating internal processes, external relationships and adaptive

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capabilities, holistic integrators are better positioned to anticipate disruptions, mobilize resources effectively and reconfigure activities in response to changing conditions. This integrated approach supports both operational efficiency and strategic effectiveness, enabling firms to sustain resilience over time. Moreover, the continuous alignment of functional, structural and strategic dimensions facilitates organizational learning and capability renewal, strengthening firms' capacity to cope with complex and prolonged disruptions.

### *6.3 Theoretical implications*

This study contributes to supply chain resilience research by advancing a multidimensional and configurational perspective on resilience capabilities. By integrating organizational functions, supply chain levels and theoretical rationales, the findings extend prior research that has predominantly examined resilience through isolated practices, functional areas or conceptual lenses. The 3D-SCRES framework demonstrates how operational, relational and transformational capabilities interact to shape resilience outcomes, thereby emphasizing the systemic and interdependent nature of resilience development.

The findings further refine the application of the RBV, RV and DCV in the context of supply chain resilience. Whereas previous studies have often applied these perspectives independently, this study illustrates how internal resource utilization, inter-organizational relationships and adaptive reconfiguration processes jointly contribute to resilience. This integrated application advances theoretical understanding of how competitive advantages and resilience co-evolve in dynamic supply chain environments.

In addition, the proposed typology of resilience orientations extends configurational and capability-based perspectives by demonstrating how different patterns of capability alignment influence firms' capacity to sustain resilience over time. The distinction between single-dimensional, partially integrated and holistically integrated orientations provides a theoretical basis for explaining heterogeneity in resilience strategies and outcomes across firms.

Furthermore, the findings highlight the context-dependent nature of resilience-enhancing activities (Fenton and Langley, 2011), contributing to ongoing debates regarding the transferability and generalizability of resilience practices. Rather than supporting universal prescriptions, this study emphasizes the importance of adaptive capability configurations that reflect organizational, structural and environmental conditions. Together, these insights position supply chain resilience as an emergent and dynamic capability grounded in the interaction of resources, relationships and reconfiguration processes.

Overall, this study makes three main theoretical contributions. First, it consolidates fragmented resilience research into a structured capability architecture spanning organizational functions, supply chain levels and theoretical perspectives. Second, it introduces the 3D-SCRES framework as an integrative lens for analyzing how resilience capabilities are enacted and aligned. Third, it develops a typology of resilience orientations that explains heterogeneity in firms' resilience strategies and outcomes.

While the 3D-SCRES framework provides an integrative perspective on how resilience capabilities are configured and aligned, its application is likely to vary across organizational and institutional contexts. Differences in firm size, governance structures, supply chain complexity and regulatory environments may influence which dimensions of resilience are emphasized and how capabilities are enacted in practice. Accordingly, the framework should be viewed as a flexible analytical lens rather than a prescriptive model. Future research may further examine how such boundary conditions shape the relevance and effectiveness of different resilience configurations.

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### *6.4 Practical implications*

From a managerial perspective, the findings suggest that supply chain resilience should be approached as a coordinated and multidimensional strategic priority rather than as a collection of isolated initiatives. The 3D-SCRES framework provides managers with a structured basis for assessing and developing resilience by explicitly considering who is responsible for resilience-related activities, where in the supply chain these activities are targeted and why they are strategically justified. This perspective supports more coherent decision-making and facilitates alignment between operational practices, inter-organizational structures and strategic objectives.

The results further indicate that managers must carefully balance efficiency-oriented practices, such as lean operations and process streamlining, with the need to maintain adequate redundancies. While efficiency initiatives may reduce short-term costs, excessive elimination of buffers and slack resources can increase vulnerability to disruptions. By adopting a multidimensional perspective, firms can identify opportunities to combine efficiency-enhancing practices, such as outsourcing, third-party logistics utilization and efficient input management, with access to redundant resources that support adaptive capacity.

In addition, the typology of resilience orientations enables firms to diagnose their current approach to resilience and identify areas for improvement. Firms characterized by single-dimensional or partially integrated orientations may benefit from strengthening coordination across organizational functions, extending resilience initiatives across supply chain levels and clarifying the strategic rationale underlying capability development. In contrast, firms pursuing holistic integration are better positioned to sustain resilience through continuous alignment and learning. Within this process, the 3D-SCRES framework can be used as a diagnostic and planning instrument to systematically map existing resilience initiatives, identify gaps across functions and supply chain levels and prioritize capability development efforts. Such structured assessments can support organizational learning and continuous improvement over time.

Finally, the findings emphasize that resilience-enhancing activities are highly context dependent. Managers should, therefore, avoid adopting standardized "best practices" and instead tailor resilience strategies to their specific organizational, network and environmental conditions. By systematically aligning capabilities across functions, partners and strategic objectives, firms can strengthen their ability to anticipate, respond to and recover from disruptions in global supply chains.

## 7. Conclusions and future research

This chapter concludes this study by synthesizing the main findings and contributions, reflecting on this study's limitations and outlining directions for future research. Building on the preceding analysis and discussion, the chapter consolidates the insights developed throughout the paper and situates them within the broader context of supply chain resilience research.

### *7.1 Conclusions*

This study synthesized the literature on supply chain resilience by examining the capabilities that firms enact to enhance readiness, response and recovery before, during and after disruptions. The findings demonstrate that resilience is not generated through isolated practices, but through coordinated constellations of activities spanning multiple supply chain levels and organizational functions. By integrating perspectives from the resource-based view, relational view and dynamic capabilities view, this research advances a

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multidimensional understanding of how resilience capabilities are developed and enacted. In doing so, this study moves beyond descriptive reviews of resilience practices and advances a configurational and capability-based understanding of how resilience is systematically developed in supply chains.

Through the application of the 3D-SCRES framework, this study clarifies where in the supply chain resilience capabilities are targeted, who is responsible for their enactment and why they are strategically justified. This capability-oriented perspective complements existing models, such as the recovery triangle (Tukamuhabwa *et al.*, 2015), by emphasizing the role of concrete organizational, relational and adaptive capabilities in shaping resilience outcomes. The proposed framework and typology provide a foundation for cumulative theory development in supply chain resilience research and support further integration of capability-based and configurational perspectives.

The findings further underscore that resilience is inherently context dependent and emerges through interactions within firms, across supply chains and within broader ecosystems. This layered perspective highlights that resilience cannot be fully understood at a single level of analysis but must instead be conceptualized as an emergent capability grounded in cross-functional coordination, inter-organizational collaboration and strategic alignment. Accordingly, resilience is best understood as a dynamic and evolving process of capability development rather than a static organizational attribute.

Taken together, the findings respond to the three research questions guiding this study. Regarding *RQ1*, the review identifies a comprehensive set of operational, relational and transformational capabilities through which firms enhance readiness, response and recovery. Addressing *RQ2*, the analysis demonstrates how these capabilities are distributed and enacted across organizational functions, supply chain levels and strategic orientations. In response to *RQ3*, this study shows that sustained resilience depends on how firms combine and align these capabilities through coherent configurations, as captured by the 3D-SCRES framework and the proposed typology of resilience orientations. Collectively, these insights highlight that supply chain resilience is not achieved through isolated practices, but through systematic capability alignment across organizational and inter-organizational contexts.

### *7.2 Limitations*

Several limitations should be acknowledged. First, this study is based on a systematic review of peer-reviewed journal articles retrieved from selected academic databases. Although established procedures were applied, relevant studies published in other outlets, languages or practitioner-oriented sources may not have been captured. As a result, the synthesis reflects the scope and structure of available academic literature.

Second, the analysis relies on secondary data, and the identification and interpretation of capabilities and activities depend on how empirical findings are reported in the original studies. Variations in research design, terminology and analytical focus across the reviewed articles may have influenced the coding and categorization process. Despite systematic procedures, thematic analysis inevitably involves elements of researcher judgment.

Third, the proposed framework and typology are derived from synthesized literature rather than primary empirical investigation. While this enables theoretical integration and conceptual development, the findings would benefit from further empirical validation across different organizational and institutional settings.

### *7.3 Directions for future research*

Future research may extend this study in several directions. First, additional systematic reviews and meta-analyses incorporating a wider range of databases, publication outlets and

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methodological approaches could further refine and expand the identified capability set. Second, qualitative and quantitative empirical studies are needed to examine how firms enact and align resilience capabilities in practice and how these configurations evolve over time.

Third, the 3D-SCRES framework and the proposed typology of resilience orientations provide a foundation for comparative and longitudinal research. Future studies may investigate how different resilience orientations influence performance outcomes and how firms transition between orientations in response to changing environmental conditions. Research examining the interactions between firm-, supply chain- and ecosystem-level capabilities would further deepen understanding of how resilience emerges through multi-level coordination.

Finally, future research may focus on operationalizing the 3D-SCRES framework through the development of measurement instruments, maturity models and assessment scales capturing the alignment of resilience capabilities across organizational functions, supply chain levels and strategic orientations. Integrating such tools with quantitative models, such as the recovery triangle, would enable systematic empirical testing, benchmarking and comparative analysis across firms and industries. This would strengthen the cumulative development of resilience research and enhance the integration of conceptual and empirical work.

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